ELEC 221 Lecture 18
CT/DT conversion and sampling DT signals

Thursday 12 November 2026

Announcements

- Quiz 8 on Tuesday (L17 and L18)
- Tutorial assignment 4 in Monday's tutorial (video processing)

- Midterm grades available

Last time

We modeled **sampling** of CT signals as multiplication of a (band-limited) signal with a periodic impulse train:

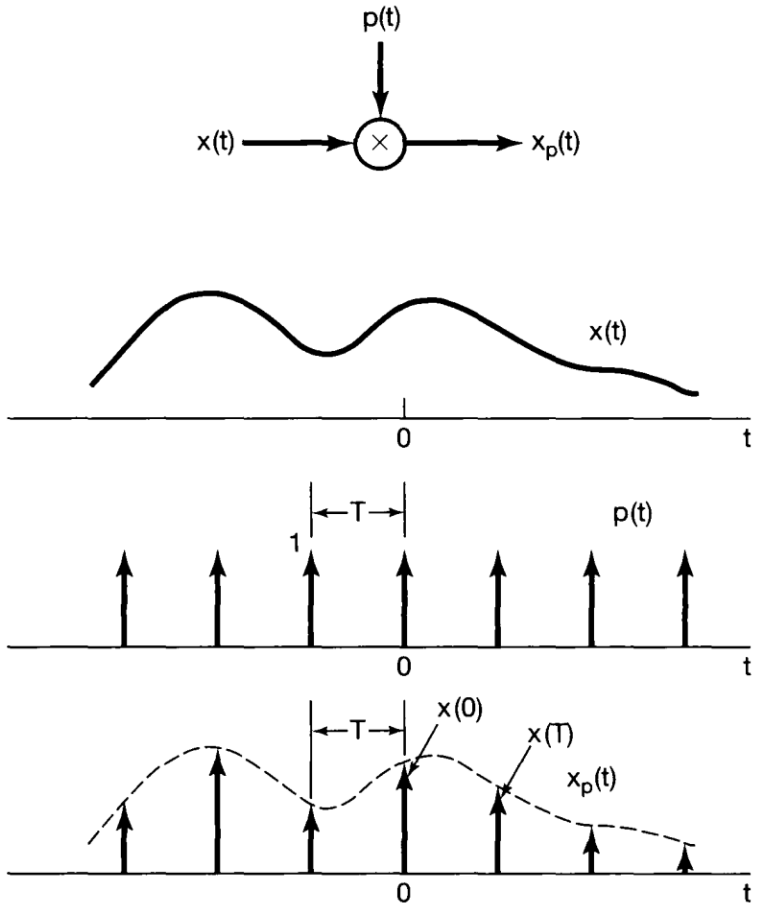


Image credit: Oppenheim 7.1

Last time

We went to the frequency domain to get a better understanding:

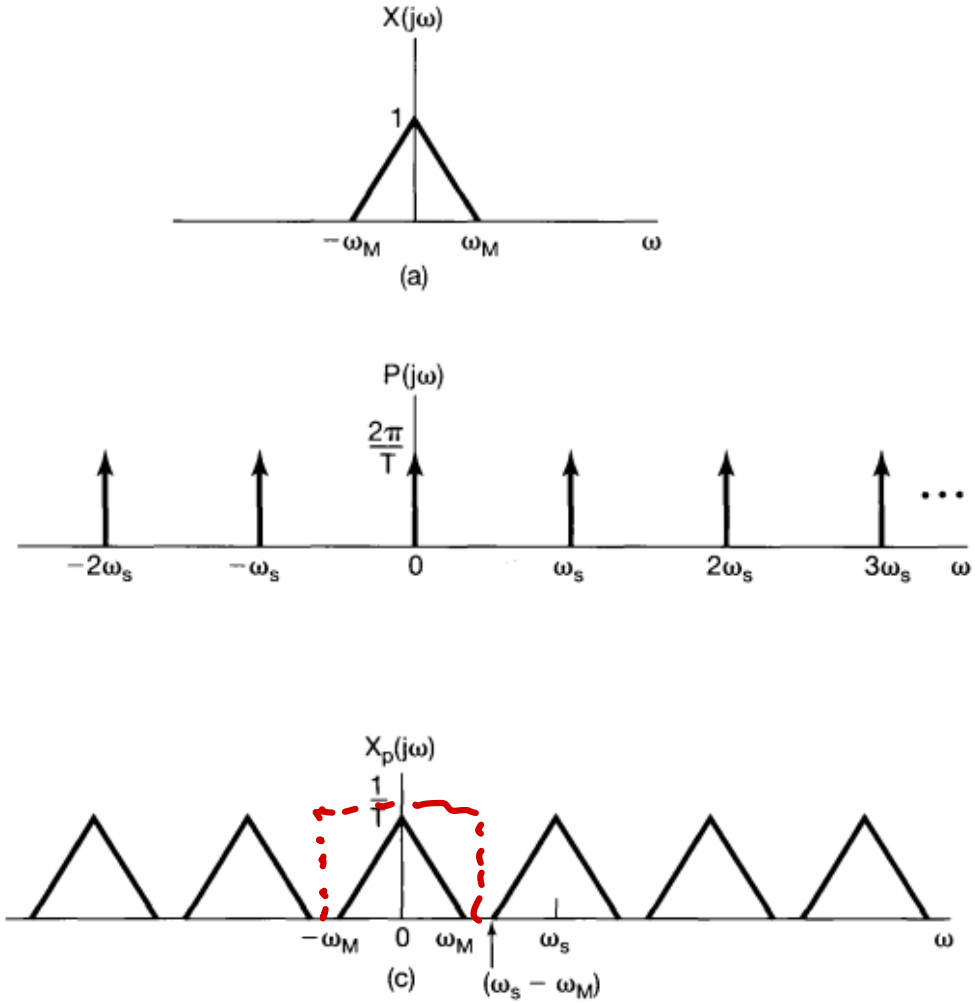


Image credit: Oppenheim 7.1

Last time

We recovered the original signal by applying a low pass filter

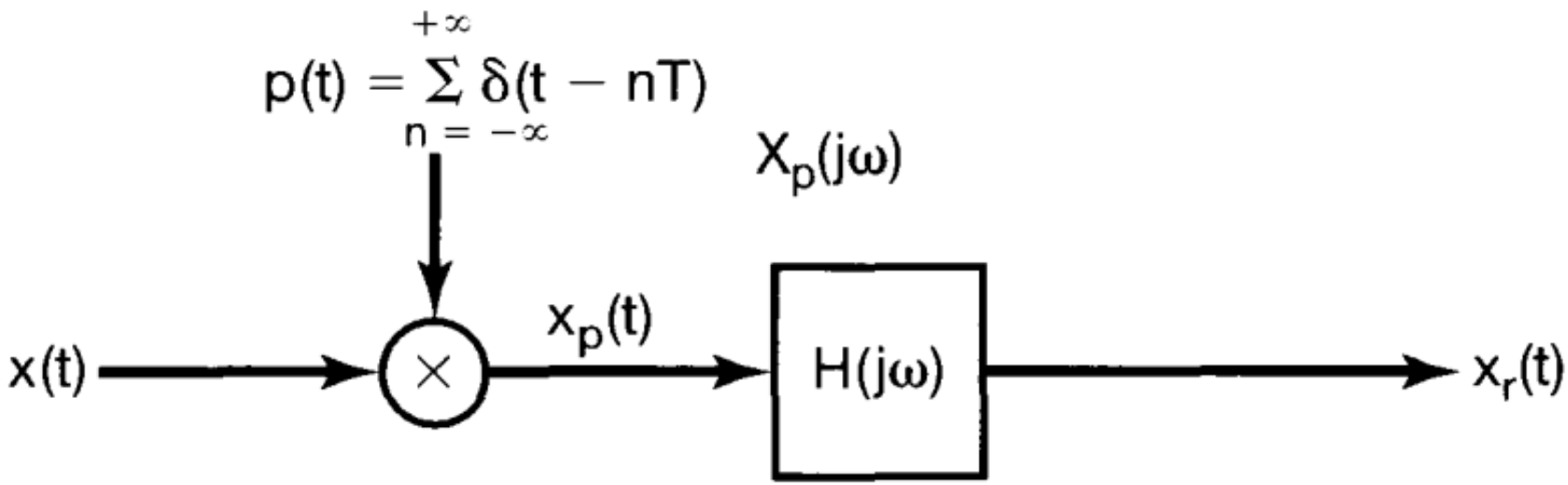
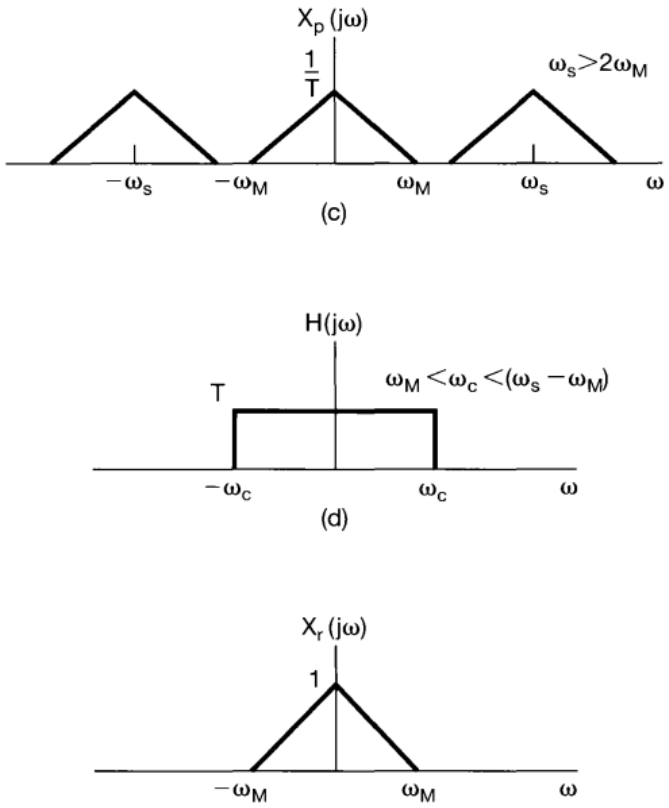


Image credit: Oppenheim 7.1

Last time



This all works if the sampling rate is higher than the **Nyquist rate**, i.e., $\omega_s > 2\omega_m$ (formally: the sampling theorem).

Image credit: Oppenheim 7.1

Today

Learning outcomes:

- describe the phenomenon of aliasing
- convert samples of CT signals to DT signals and analyze the frequency domain effects
- perform impulse train sampling on a DT signal
- decimate and interpolate a DT signal, and describe their effects on the spectrum

Aliasing

What if the sampling rate isn't high enough?

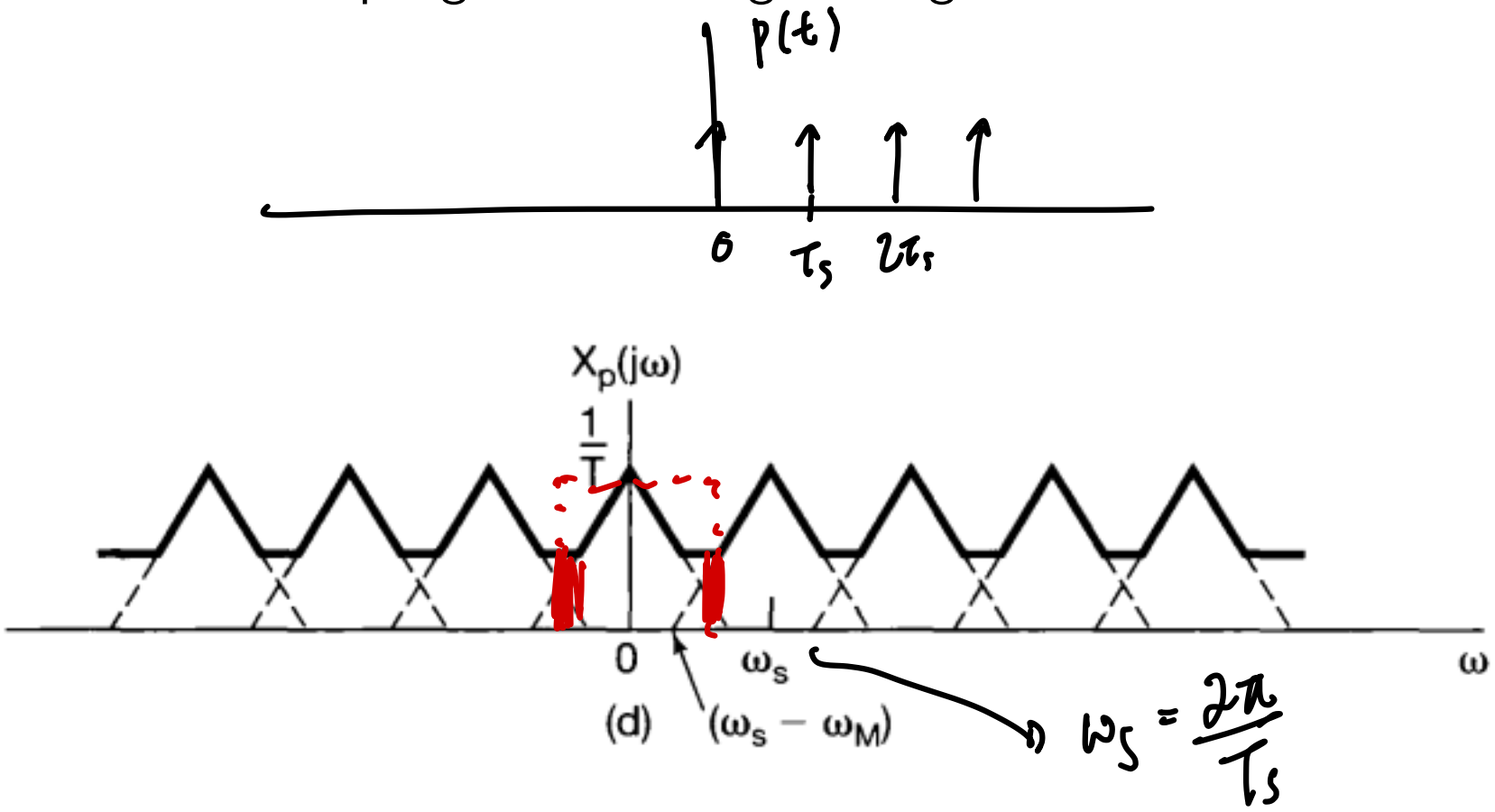


Image credit: Oppenheim 7.1

Aliasing

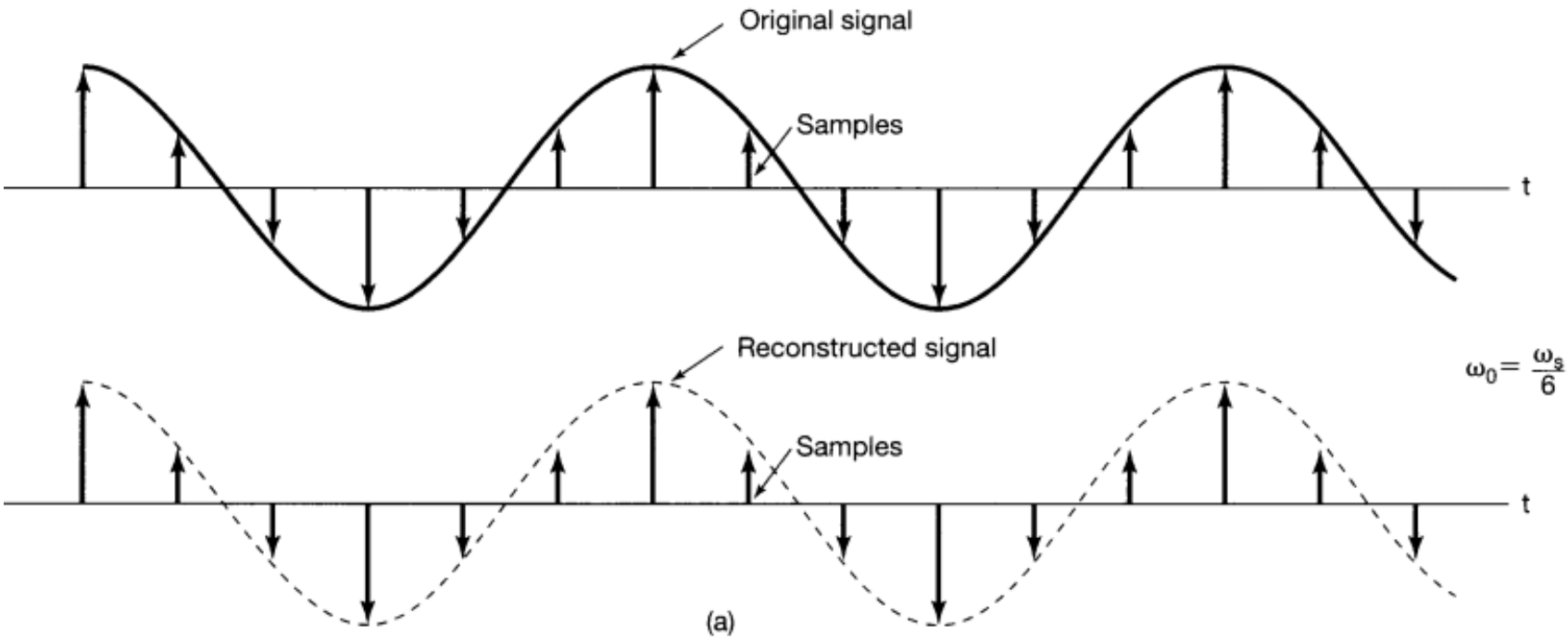


Image credit: Oppenheim 7.3

Aliasing

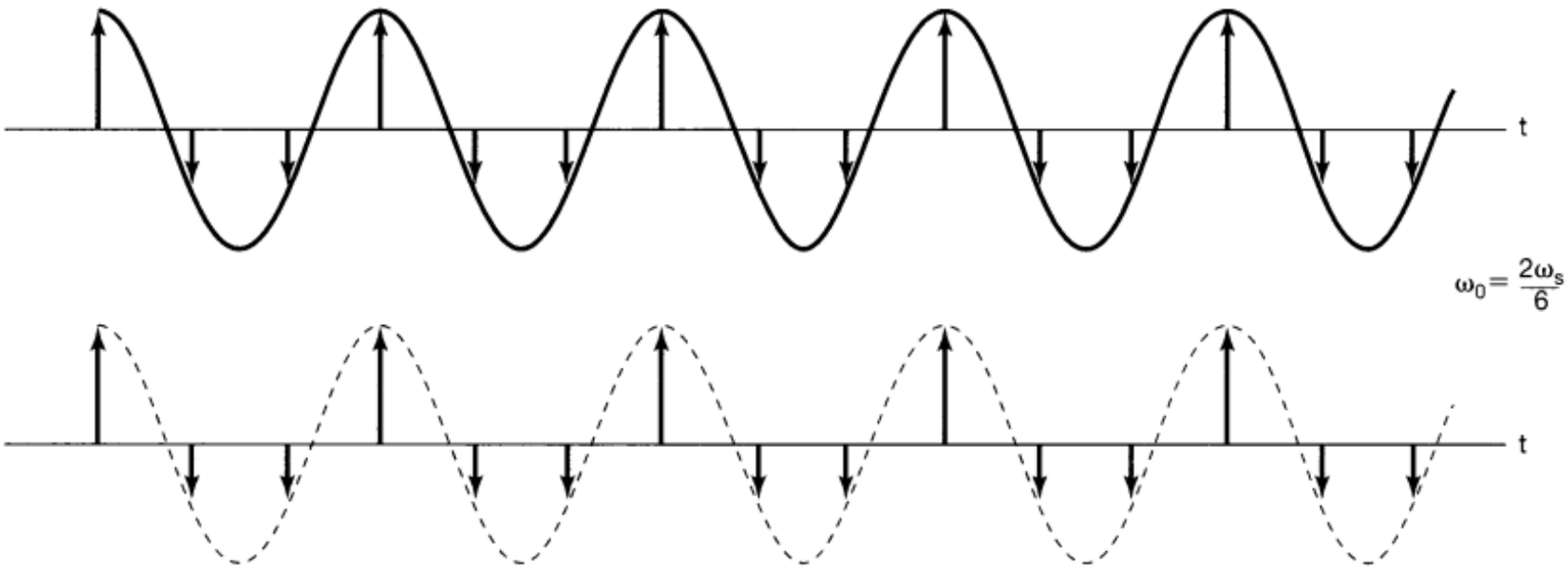


Image credit: Oppenheim 7.3

Aliasing

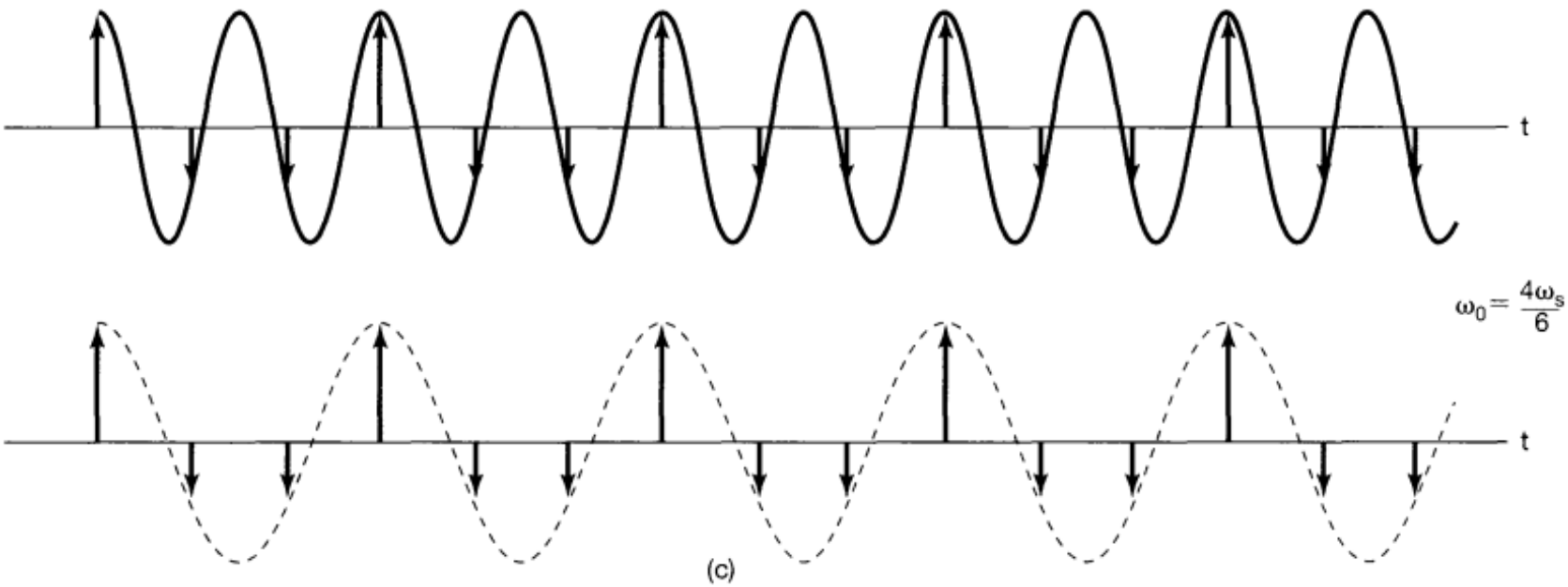


Image credit: Oppenheim 7.3

Aliasing

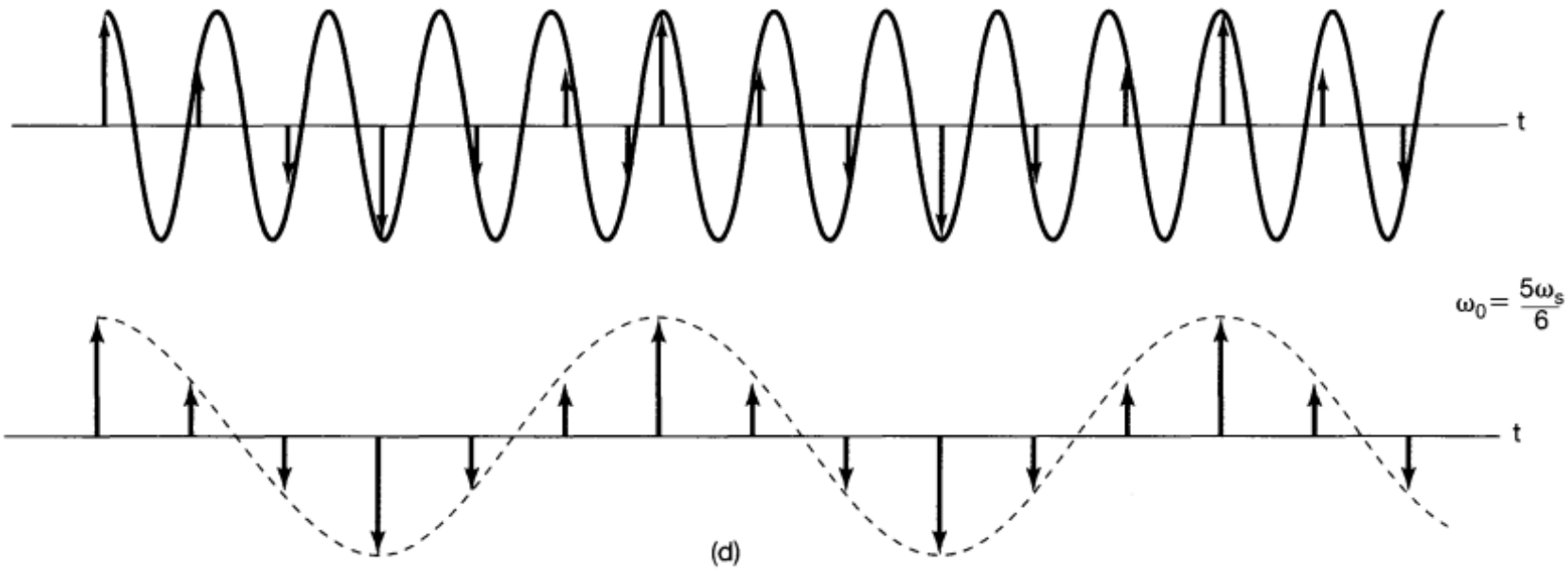


Image credit: Oppenheim 7.3

Simulation

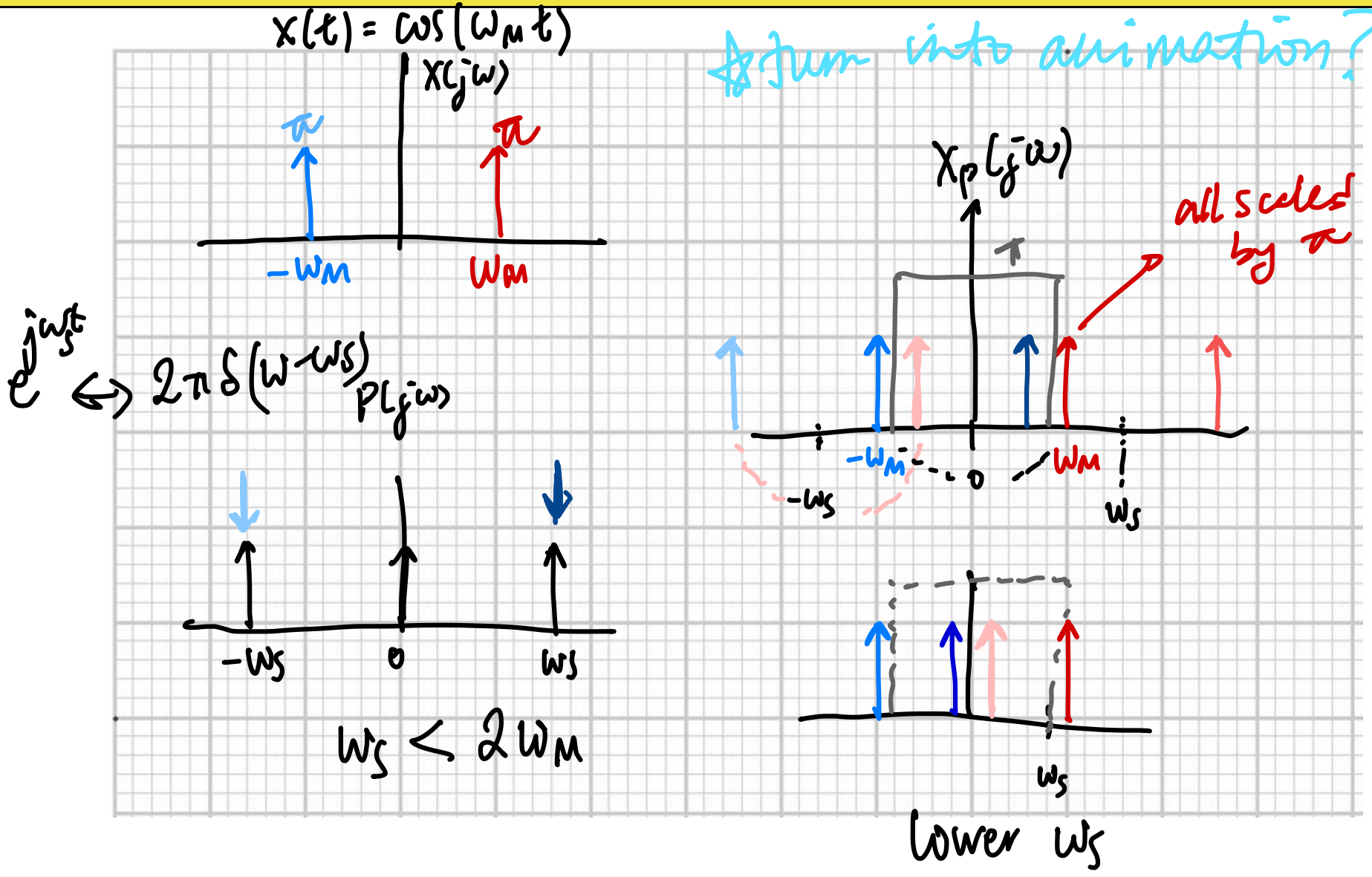
`https://visualize-it.github.io/stroboscopic_effect/simulation.html`

Two aspects to consider here:

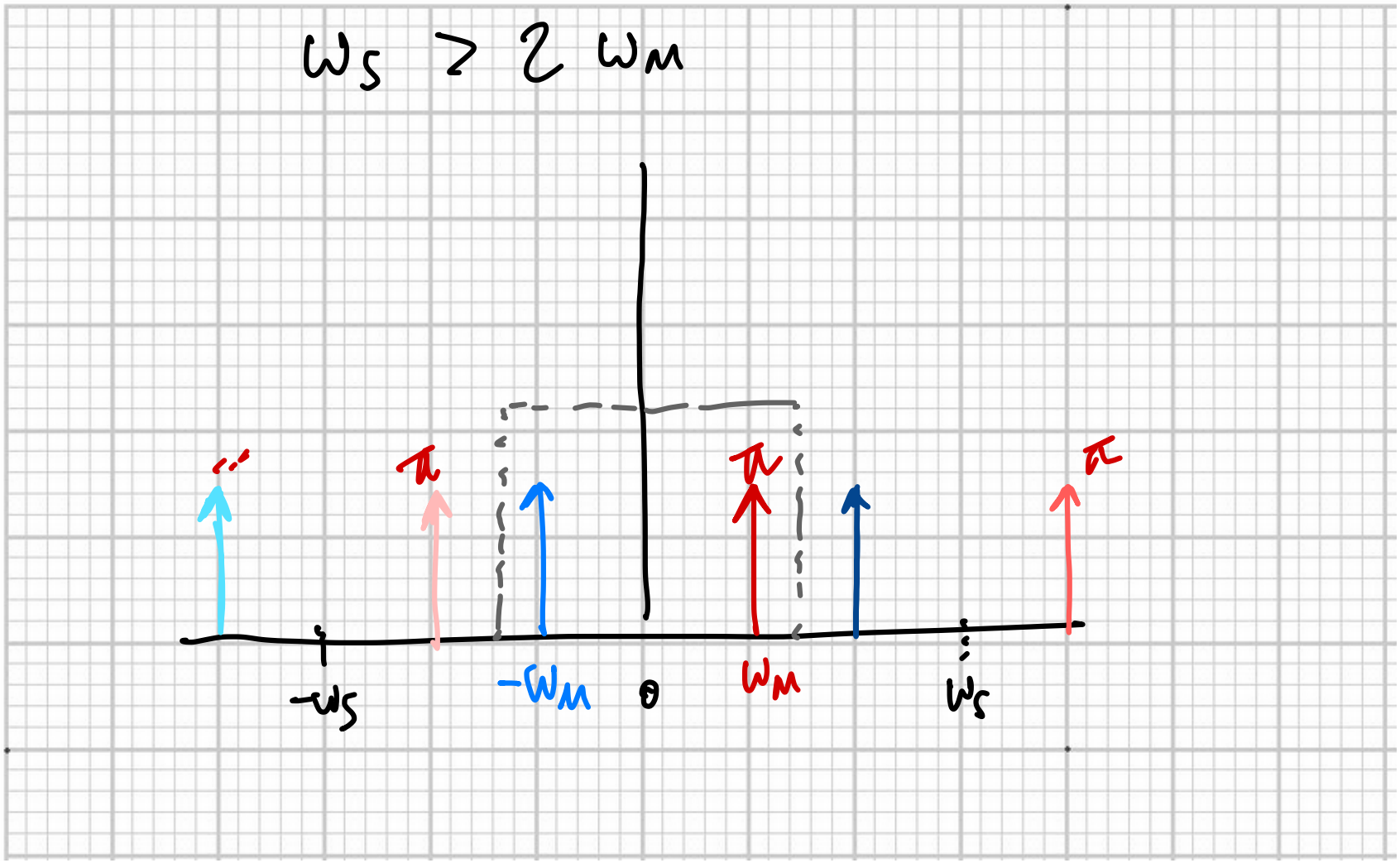
- Why does the interpreted frequency *decrease* as the true frequency increases?
- Why does it it look like it goes *backwards*?

We can understand both by looking at the spectra.

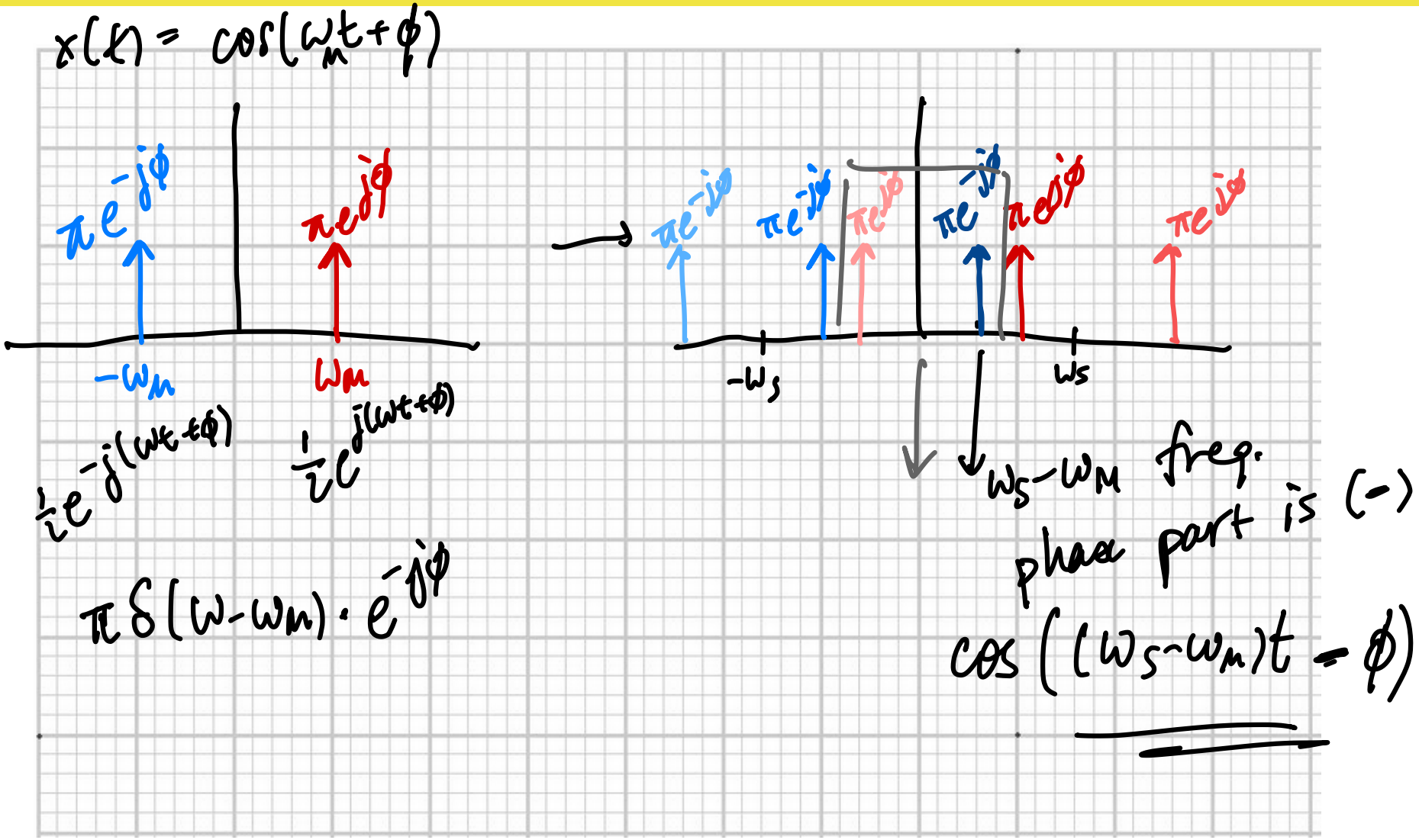
Frequency misattribution



Frequency misattribution

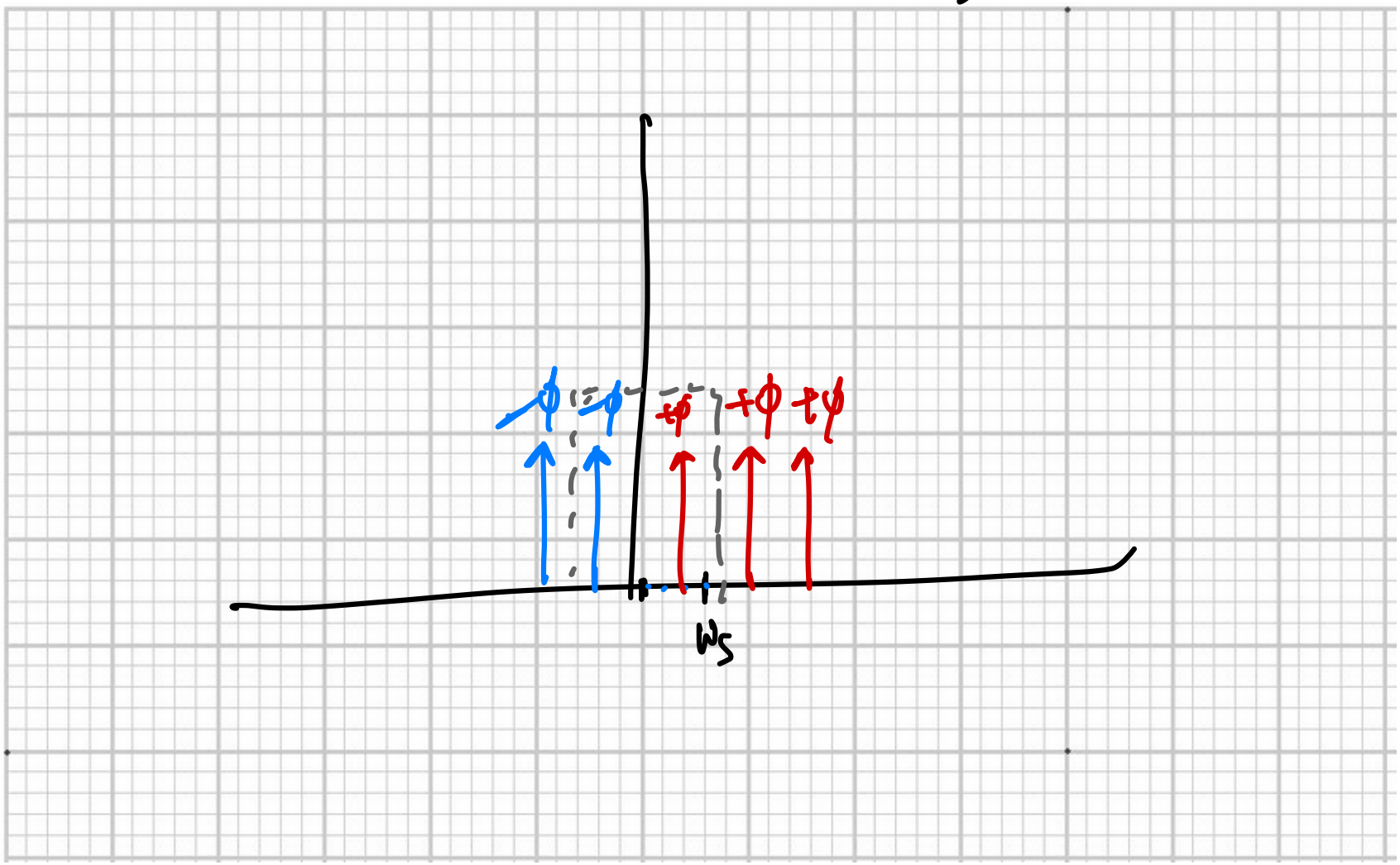


Backwards-ness

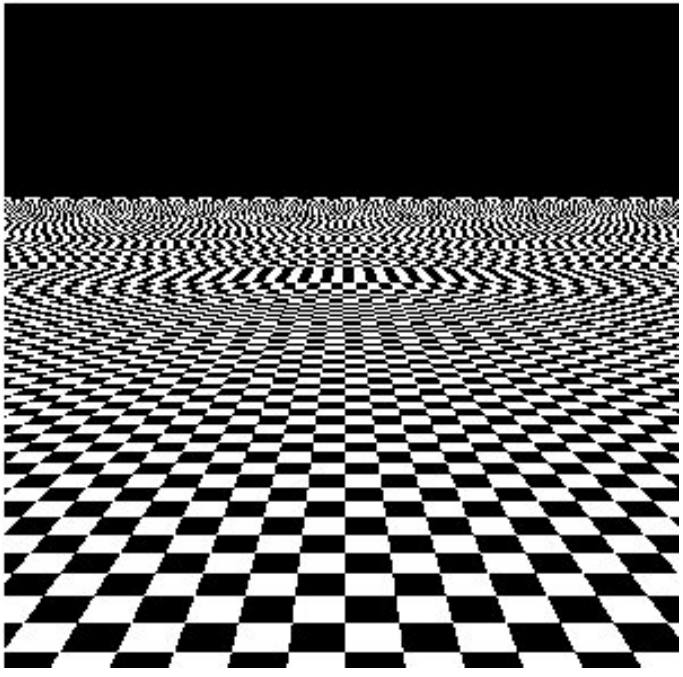


Backwards-ness

$\omega_s < \omega_N$



Real-world examples



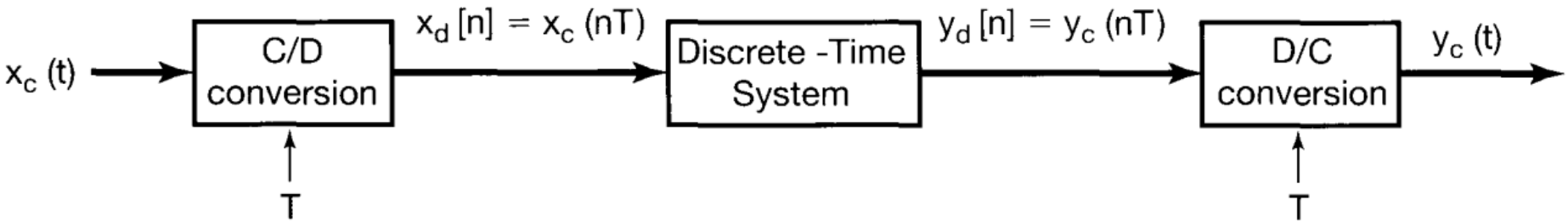
Fun on your own: read up about Moiré patterns, and various **anti-aliasing** techniques that are used in music/images/games!

Image credit: `https:`

`//textureingraphics.wordpress.com/what-is-texture-mapping/anti-aliasing-problem-and-mipmapping/`

Converting between DT $\leftrightarrow$ CT

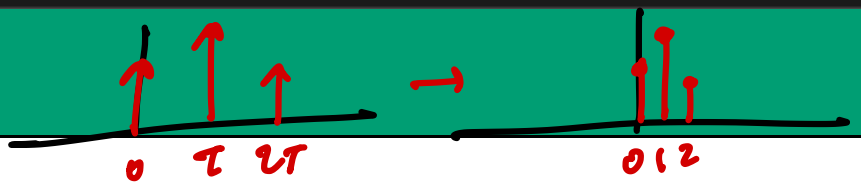
Often convenient to process CT signals by first converting to DT, processing, then converting back.

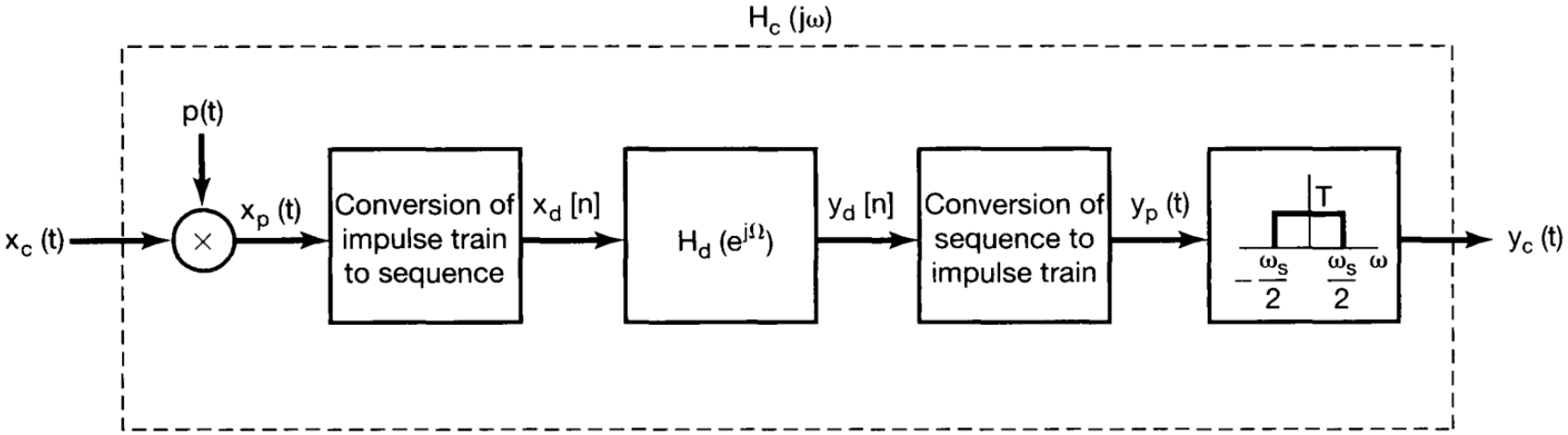


What is the theory that makes this possible?

Image credit: Oppenheim 7.20

Converting between DT $\leftrightarrow$ CT





Let's explore what happens at the level of the spectra

Note: we have *two frequencies*, one in CT, one in DT. Write:

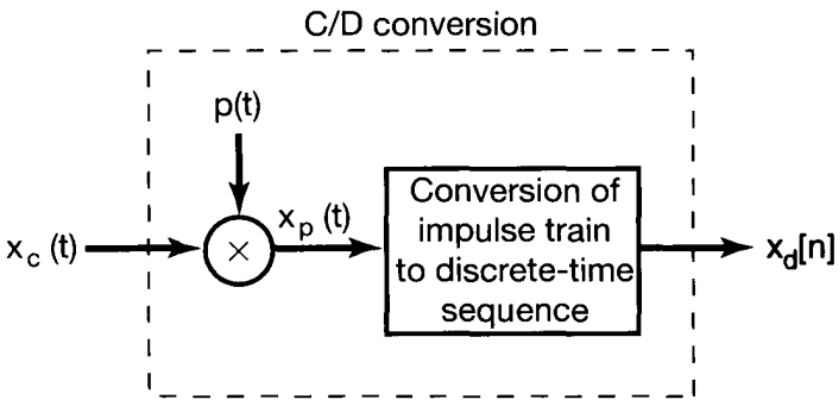
$$X_c(j\omega), \quad Y_c(j\omega)$$

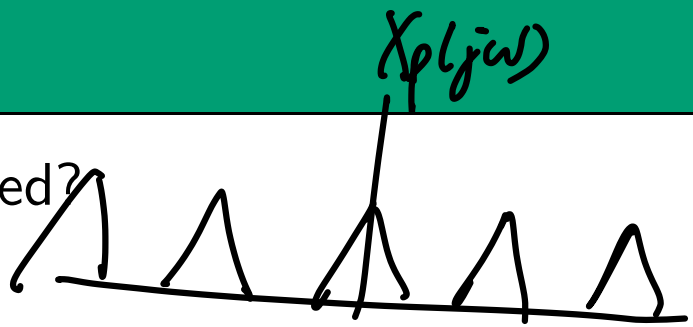
$$X_d(e^{j\Omega}), \quad Y_d(e^{j\Omega})$$

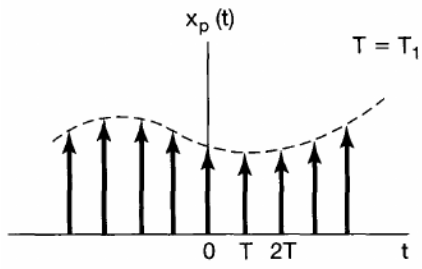
Image credit: Oppenheim 7.24

Converting between DT ↔ CT

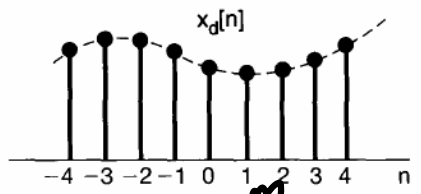
First: how are $X_p(j\omega)$ and $X_d(e^{j\Omega})$ related?







as a Fourier series



$$x_p(t) = \sum_{k=-\infty}^{\infty} x_c(kT)\,\delta(t-kT) \longrightarrow X_p(j\omega) = \sum_{k=-\infty}^{\infty} x_c(kT)\,e^{-j\omega kT}$$

DTFT

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x_d[n]\,e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x_c(n\cdot T)\,e^{-j\Omega n} = X_p(j\Omega/T)$$

Converting between DT $\leftrightarrow$ CT

Relate $X_d(e^{j\Omega})$ back to the original spectrum $X_c(j\omega)$

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\omega - k\omega_s))$$

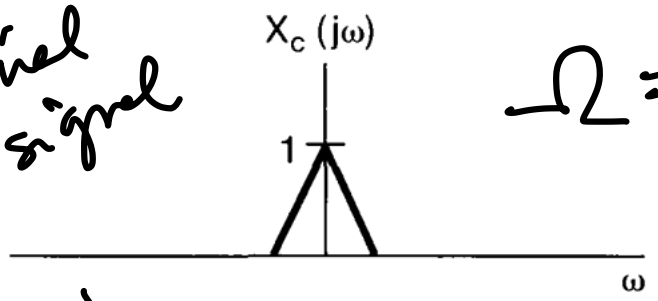
$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$

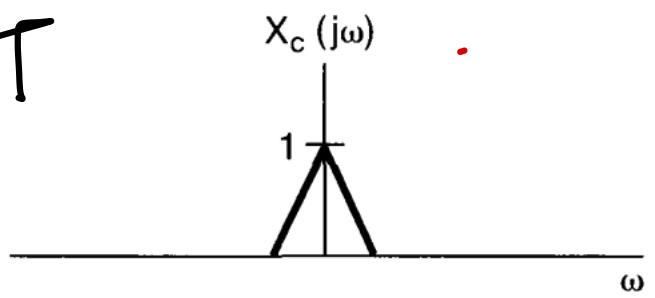
$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega/T - k\omega_s))$$

Converting between DT ↔ CT

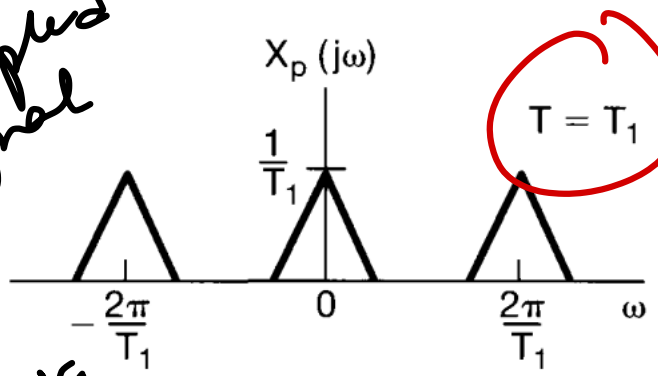
original CT signal

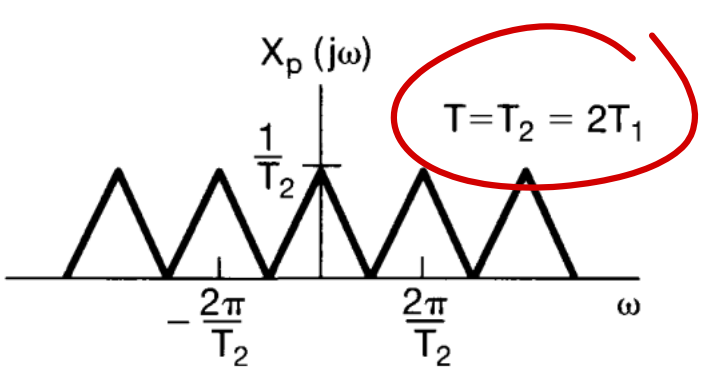
$\Omega = \omega T$



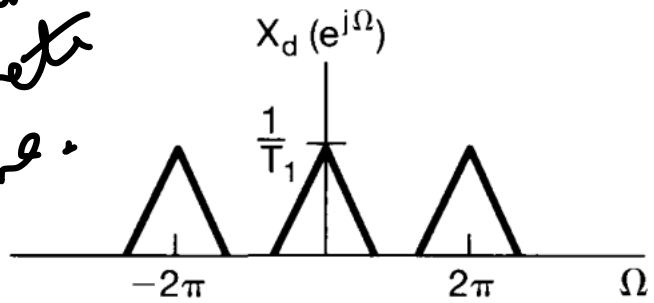


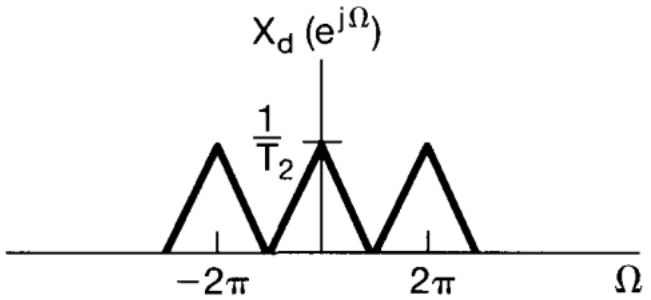
CT sampled signal



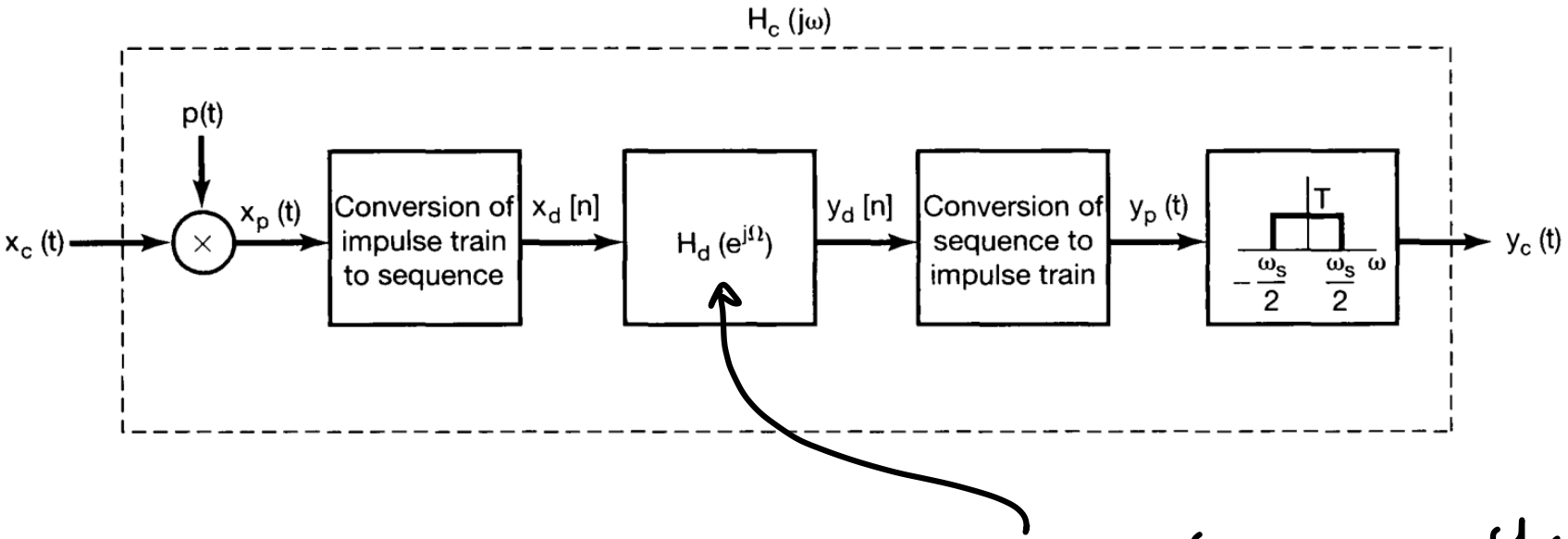


after conv. to discrete time.





Converting between DT $\leftrightarrow$ CT



The converted signal $x_d[n]$ enters a DT system: w/ freq. resp. $H_d(e^{j\Omega})$

$$Y_d(e^{j\Omega}) = H_d(e^{j\Omega}) X_d(e^{j\Omega})$$

$$= H_d(e^{j\Omega}) \cdot \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega/T - k\omega_s))$$

$$= H_d(e^{j\Omega}) \cdot \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\frac{(\Omega - 2\pi k)}{T}\right)$$

Image credit: Oppenheim 7.4

Converting between DT ↔ CT

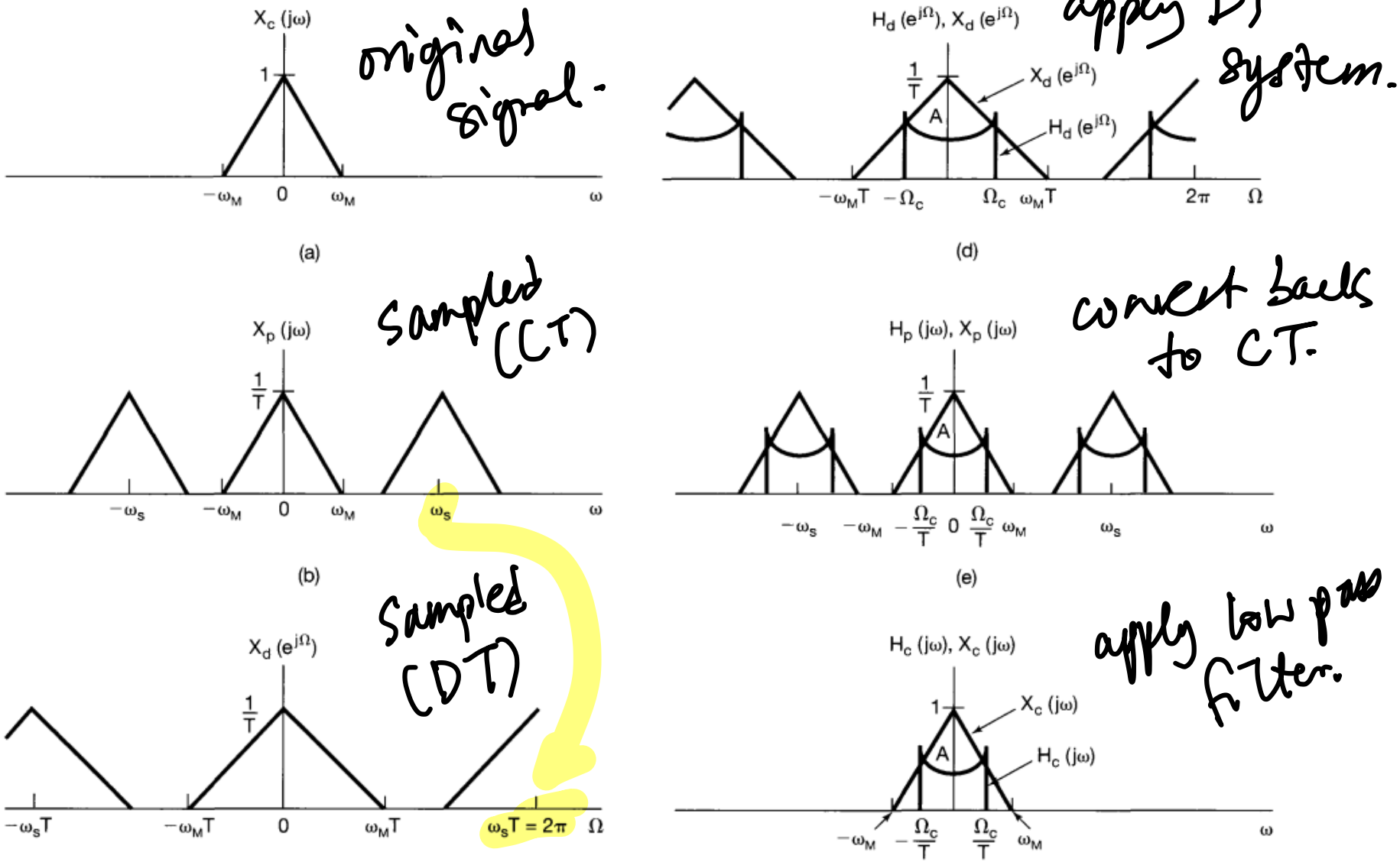


Image credit: Oppenheim 7.4

Sampling of DT signals

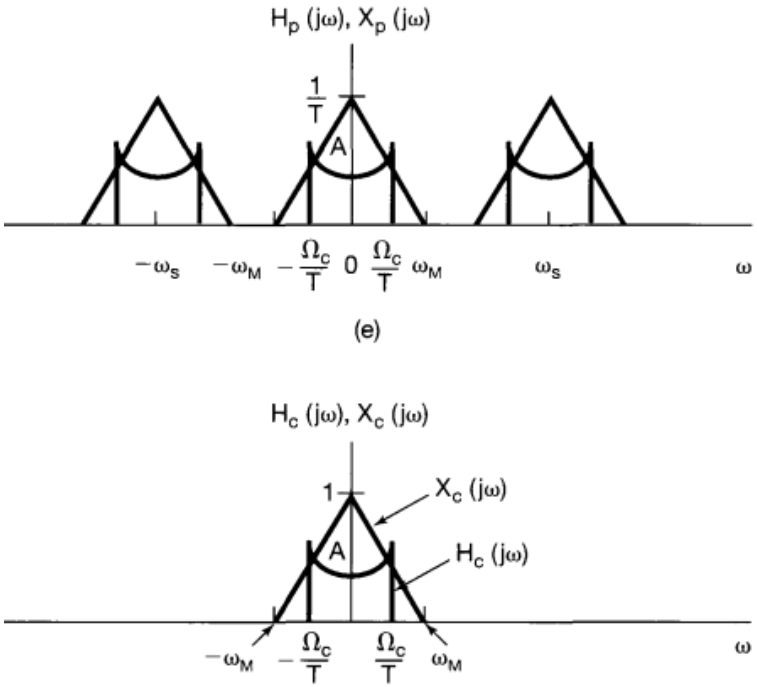
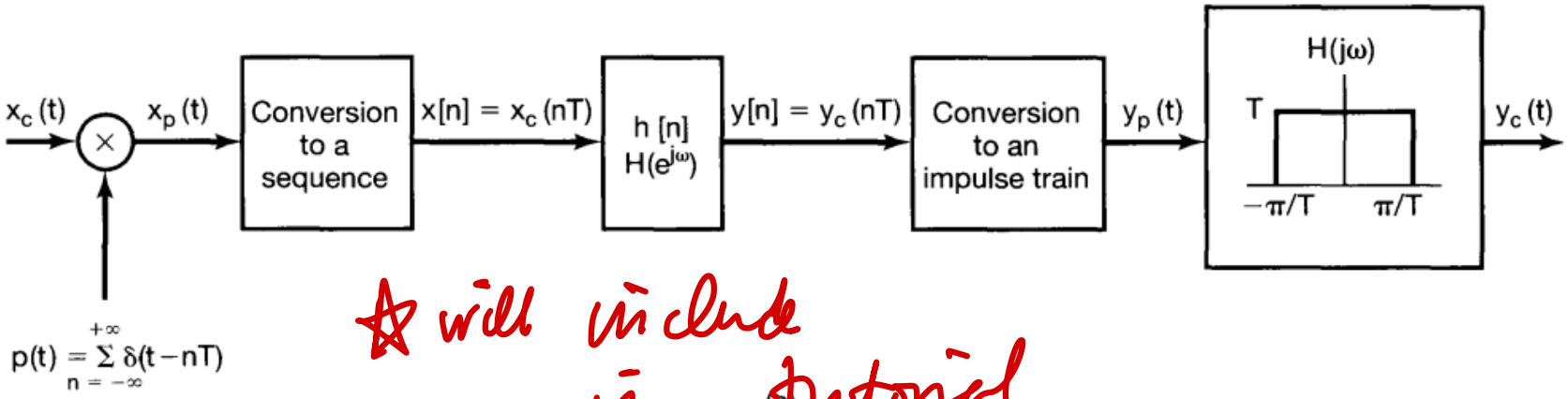
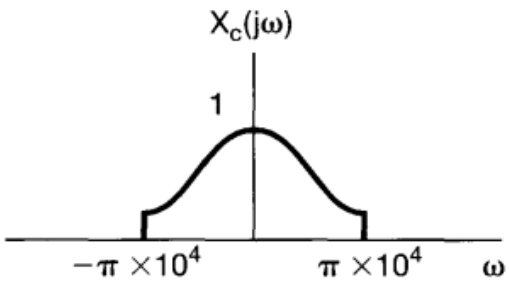


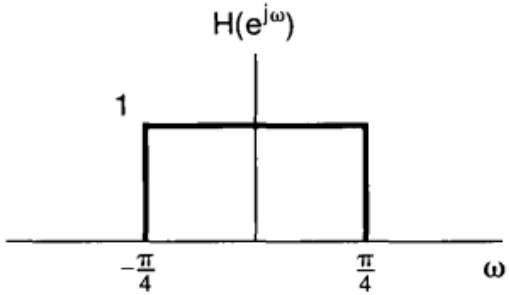
Image credit: Oppenheim 7.4

Example



will include in tutorial





Sketch: $X_p(j\omega)$, $X(e^{j\omega})$, $Y(e^{j\omega})$, $Y_p(j\omega)$, $Y_c(j\omega)$ if $1/T = 20kHz$.

Image credit: Oppenheim Problem 7.29

Example

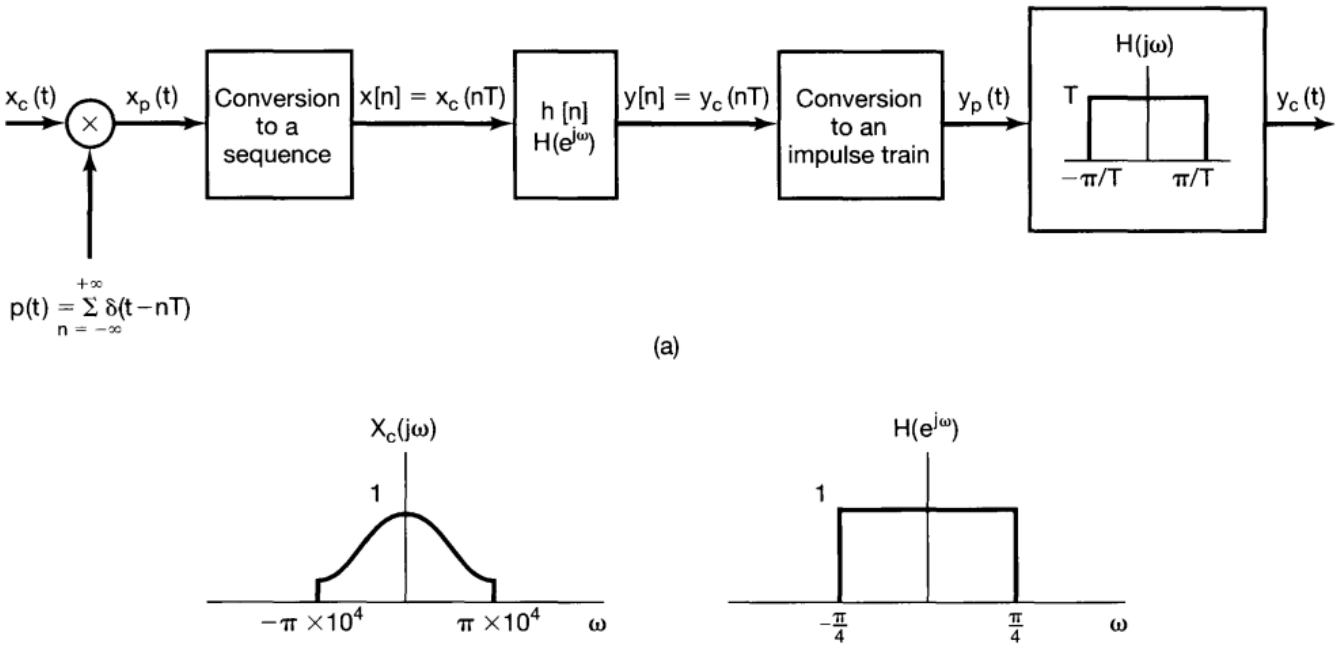


Image credit: Oppenheim Problem 7.29

Example

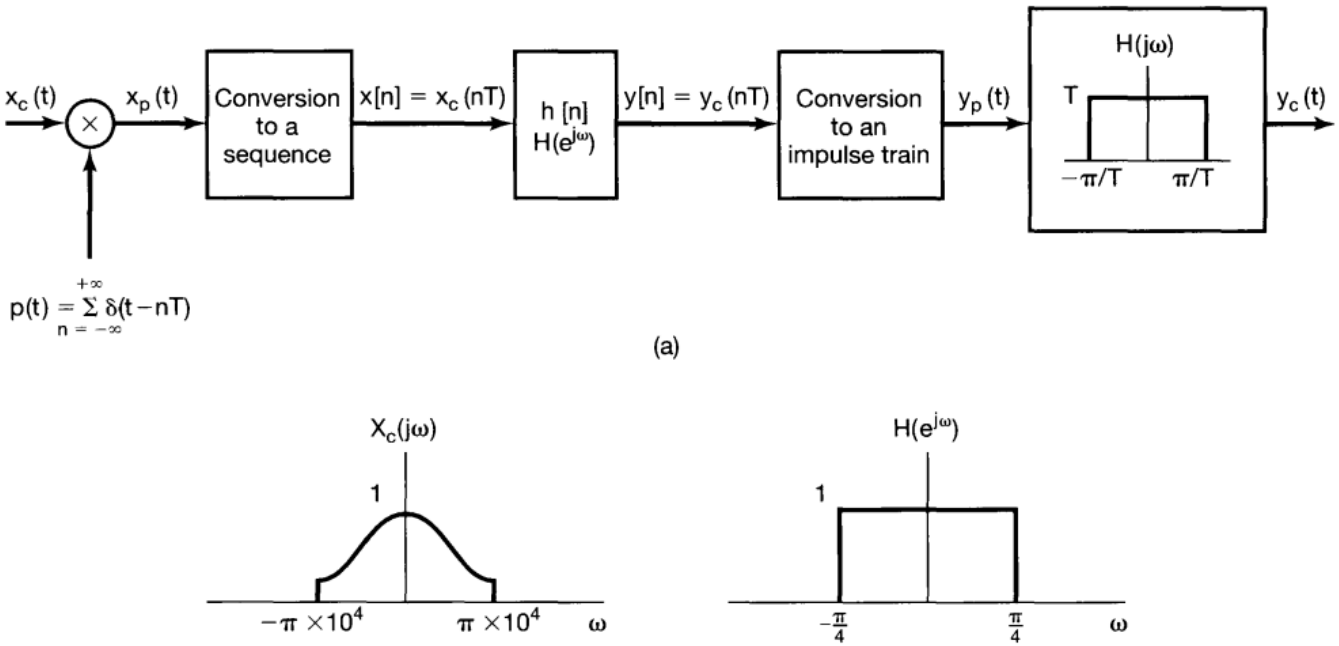


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Example

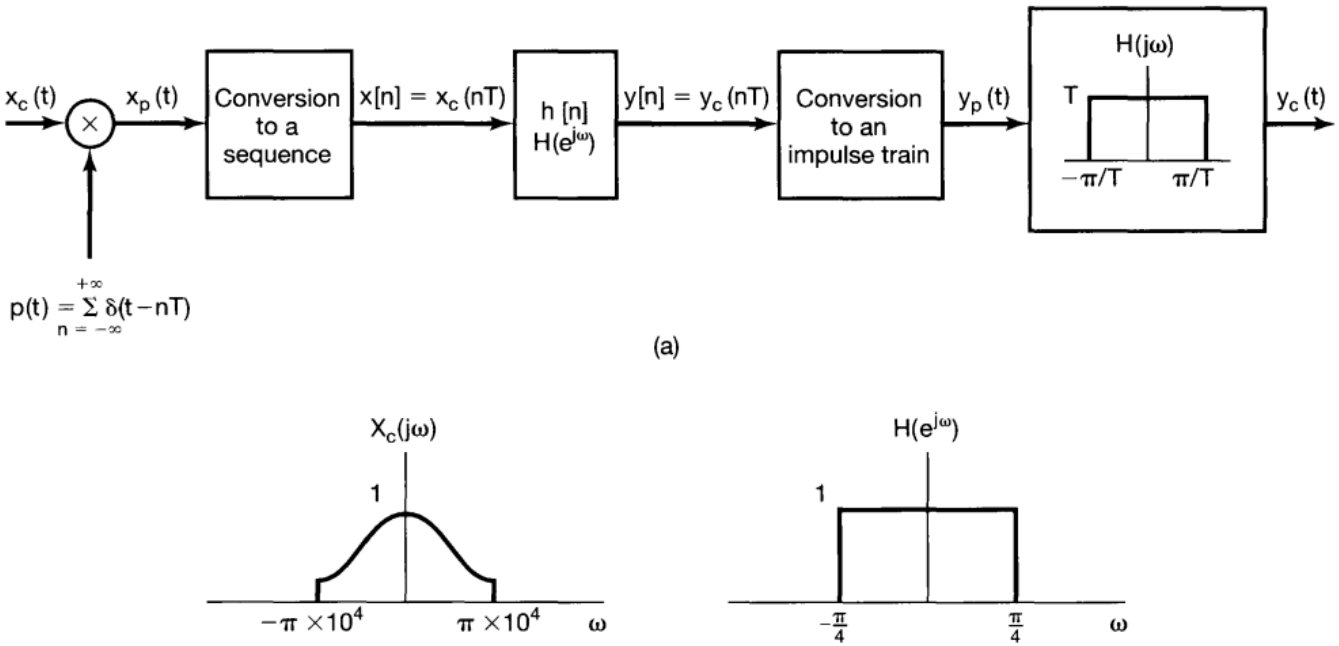


Image credit: Oppenheim Problem 7.29

For next time

Content:

- new topic: modulation / communication systems

Action items:

1. Try questions in practice assignment 4
2. Prepare for quiz 8 on Tuesday (L17 and L18 material)
3. Tutorial Assignment 4 Monday

Recommended reading:

- From this class: Oppenheim 7.4-7.6
- Suggested problems: 7.17, 7.18, 7.20, 7.30, 7.32
- For next class: Oppenheim 8.0-8.4